Boosting Education Technology in Africa

Scaling Innovation for the Future of Education

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Key Takeaways

Africa struggles to embrace the needs of its ever-growing population: 98 million children in Sub-Saharan Africa are currently out-of-school, while 72 million young people, aged from 15 to 24, are neither trained nor employed. Two thirds of them are women. In such a context, digital tools can foster transformation:

1. Scalable solutions, such as mobile platforms or AI-powered learning applications, can improve access to qualitative courses that are in line with local working imperatives. .
2. African EdTech startups are lagging, as they are battling to raise sufficient fundings. They are also facing market fragmentation and regulatory challenges.
3. Unlocking their capacity is key to progress. Governments, businesses, as well as accelerators should collaborate to deploy their continent's EdTech potential. Regional partnerships, broader investments and more flexible policies are vital.

This article explores the ways in which a vast EdTech reform can respond to Africa's pressing educational difficulties, its needs for improvement and lasting change.

Education: One of Africa's Key Challenges

Africa's population is the world's youngest, as it crosses 1.4 billion, 70% of the population is under 30 years old. By 2055, over one billion children will be living on this continent. However, Africa's education system is facing three significant challenges that need to be quickly addressed.

1 **Access:** In sub-Saharan Africa, 98 million children are out-of-school. Between 2015 and 2021, this number rose by 12 million. Girls, rural communities and low-income families are the most exposed to this phenomenon.

2 **Quality:** The continent lacks around 15 million teachers. This shortage, combined both with inadequate training and outdated curricula, generates limited learning outcomes. In some regions, less than half of teachers have an appropriate degree, while absenteeism is very high.

3 **Skill Gaps:** With 72 million young people (15-24) neither in education nor employment, the skills mismatch hinders economic progress, particularly for women, who make up two-thirds of this group.

These figures talk for themselves:
A large-scale education reform is vital. Digital solutions are essential.

Fast-Tracking Education Change with Digital Solutions

Not only can digital technologies strengthen education at all levels, starting from early childhood to adult training, they are also more inclusive, as they provide scalable, cost-effective means to reach all populations.

In rural areas, mobile-based learning platforms are key to bridging skill gaps. For instance, the Kenyan platform Eneza Education has reached five million users through basic mobile phones, with consequent results: exam scores improved by 5%. In Tanzania, another platform responds to Africa's urgent need for learning systems: 17 million children use Ubongo each week, through its academic programs.

Many additional tools rely on AI to tailor content, allowing students to work at their own pace. In Brazil, Khan Academy has improved math proficiency, as it was integrated into public education.

Other startups target adult learners. For instance, Alt_School provides online courses leading to several certifications. The Egyptian platform Almentor offers several courses and contents in Arabic. Others like Andela aim to train their users to an evolving market and provide them with in-demand tech needs. They also connect them with recruiting companies. Another example is that of Sayna, a Malagasy provider of gamified digital courses in tech skills, that also connects its learners to lucrative IT micro-tasks. High school and university students can rely on Gradesmatch, an orientation platform that helps them understand the different kinds of careers they can aim at.

Technology supports students but also constitutes a valuable tool for teachers. In South Africa, AI-driven solutions enable professors to reduce their number of administrative tasks, which give them more time to focus on student engagement. As early as 2015, Rwanda adopted the Comprehensive Assessment Management Information System (CA-MIS): through this application, teachers can follow their students' progress in real time. In Nigeria, educators rely on Edves, a learning software that enables them to track academic progress.

These statistics could grow, should Africa's digitalization expand. Projections are encouraging to that effect: mobile internet penetration is expected to reach 51% in 2025, which is quite impressive, as it was only 3% in 2010.

More than ever, technology appears to be the strongest tool at Africa's disposal to promote education.

Edtech in Africa needs a boost

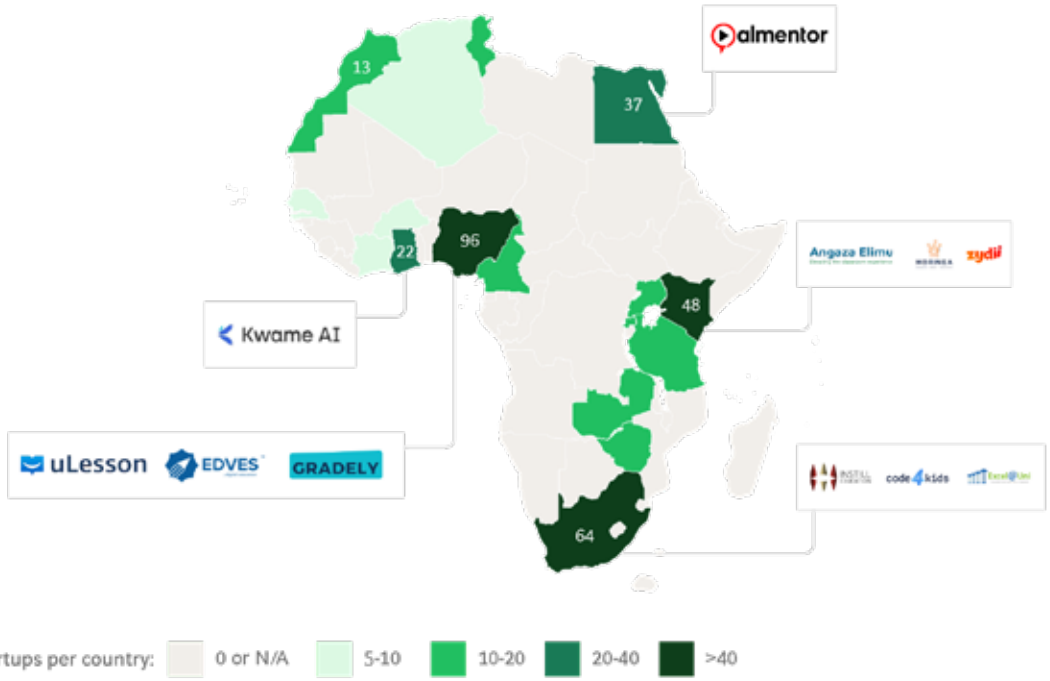
Over four hundred EdTech startups have developed throughout Africa, half of which are located in three countries: Nigeria, South Africa and Kenya.

EXHIBIT 1

Africa's EdTech Startups Landscape: Concentrated in Nigeria, South Africa and Kenya

There are ~400 EdTech startups in Africa as of beginning of 2022, mostly concentrated in Nigeria, South Africa and Kenya

Mapping of EdTech startups in Africa



Source: Briter Bridges Intelligence platform; Holon IQ; BCG Analysis

EdTech startups in Africa primarily serve **K-12** education, with around 64% targeting this segment, while fewer address higher education and professional training. In comparison, North American EdTech startups emphasize workforce training and higher education more heavily, with only about 25% focusing on K-12, about 50% targeting vocational and workforce learning, and the remainder 25% serving higher education. Asia falls between these two regions, with roughly 40% of Southeast Asia's EdTech startups concentrating on K-12, while 60% focus on vocational training and higher education.

Additionally, over 80% of these African startups provide e-learning solutions, including MOOCs, that provide scalable, unlimited participation and open access. However, these platforms often struggle to deliver personalized, engaging, and integrated learning experiences compared to other educational technologies. Furthermore, despite the sector's potential, more than 70% of EdTech companies are founded by men, highlighting a significant gender gap in leadership.

Funding continues to be the main obstacle. EdTech accounts for only 4% of African-tech fundraising. In 2023, less than $10 million were raised in leading markets like Egypt and Nigeria, which is far below the average of other emerging economies.

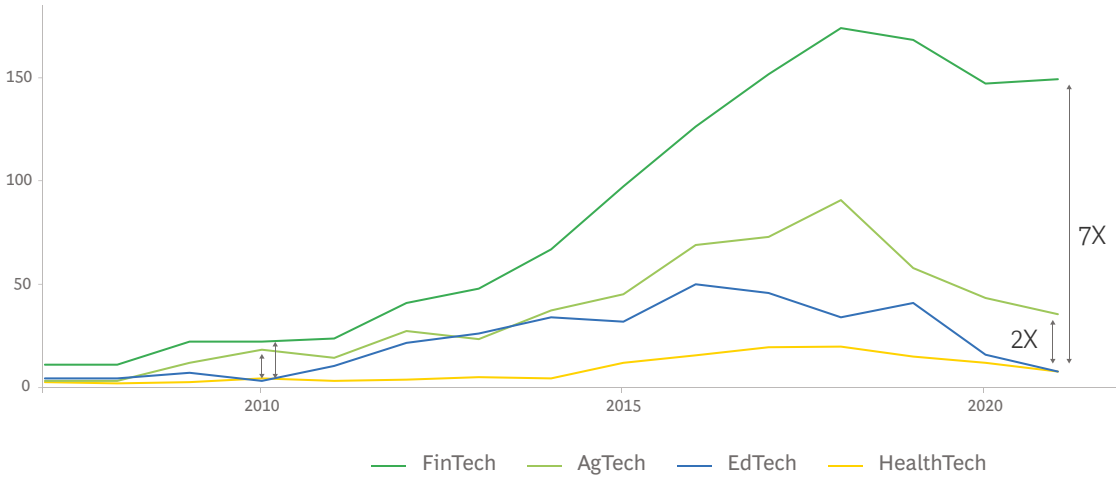
This lack of fundraising results in **limited startup development.** Despite an initial period of growth, the rate of new EdTech ventures has markedly slowed since 2019, even though the global shift toward digital solutions during the COVID-19 pandemic provided substantial opportunities in other sectors. Compared to sectors like FinTech and AgTech, EdTech's progress has notably lagged behind; between 2010 and 2021, the gap in the creation of new EdTech startups widened by a factor of seven relative to FinTech, and doubled compared to AgTech.

EXHIBIT 2

Growth of EdTech Companies Lagging Behind: The Widening Gap Since 2010

The gap between the number of startups founded between EdTech, AgTech and FinTech has grown substantially from 2010

Number of tech startups founded over the years in Africa



Between 2010, 2021, the gap in new startups in EdTech compared to FinTech widened by a factor of 7, while it doubled for AgTech

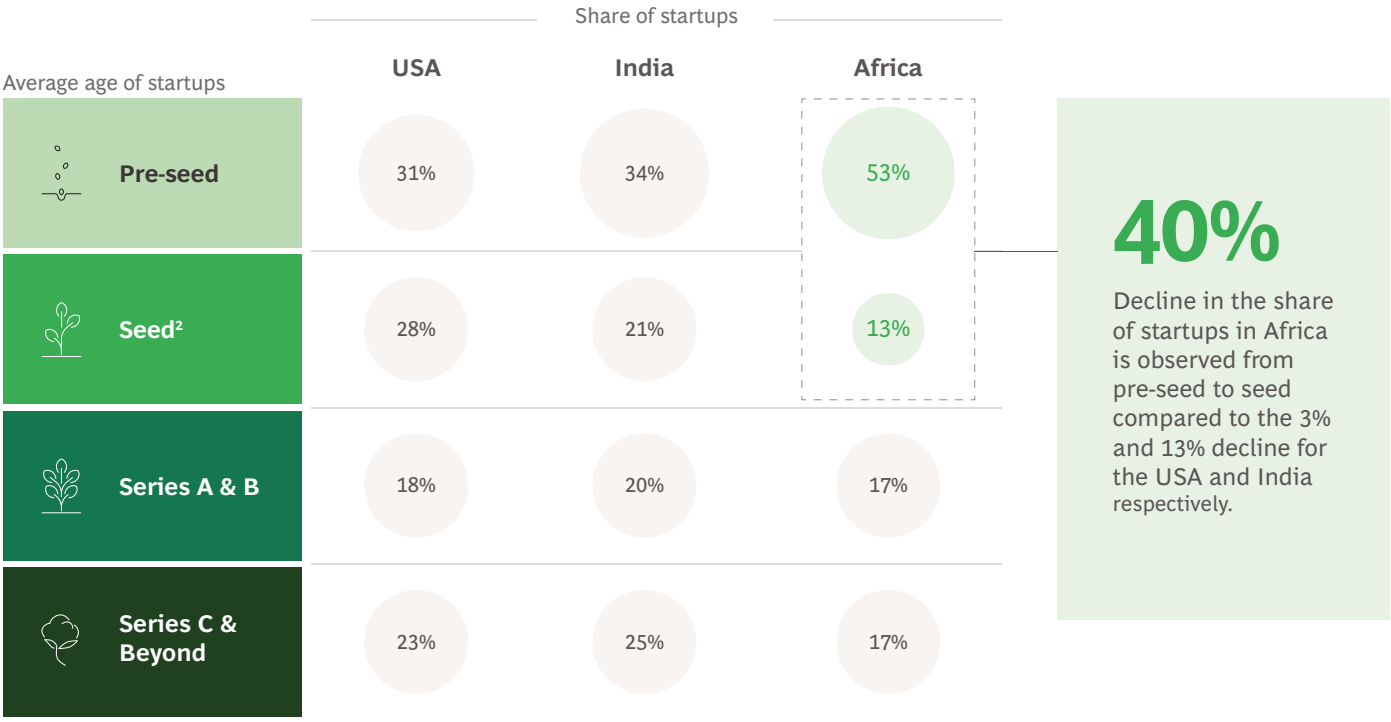
Source: Briter Intelligence, BCG Analysis

This slow progression is further compounded by the dominance of very early-stage startups in Africa. Most African EdTech startups are concentrated at the pre-seed stage (53%), with significantly fewer at the seed stage (13%) or Series A & B and beyond (34%). In contrast, the USA and India have a more balanced distribution across funding stages, highlighting stronger scaling potential.

EXHIBIT 3

Funding Stage Imbalance: African EdTech Startups are predominantly in Early Stages

EdTechs in Africa are confronted with the challenge of scaling up



1. Startups that managed to raise funds in the next round in 2023
2. Includes grant funding

Source: PitchBook, BCG Analysis

Last but not least, EdTech startups face **important market fragmentation:** both Africa's linguistic diversity and regulatory inconsistencies generate major barriers when it comes to scaling across national borders.

Nevertheless, **some success stories illustrate the potential of EdTech in Africa** if existing challenges were effectively addressed. In Nigeria, uLesson has successfully raised $26.5 million, achieving over 3.5 million app downloads as of 2023, with users viewing more than 12 million educational videos, significantly expanding access to high-quality K-12 content. In South Africa, FoondaMate's AI-powered platform, designed to deliver educational resources to remote areas with limited connectivity, reached over 3 million users by 2024.

Scaling Edtech in Africa: Four Solutions Worth Trying

Unlocking Africa's EdTech potential depends on four key aspects:

1 **Government vision and policy:** Clear domestic strategies can encourage EdTech's expansion. Policies such as active sourcing, technology grants, and subsidies create a hospitable environment. To that extent, Rwanda's ICT policy, focusing on education, sets an example: it aims to extend connectivity nationwide, thus integrating digital tools to classrooms, reinforcing teacher training and spreading student access to programs. Though recent, its impact is already here.

2 **Business and users' alignment:** Organizations and other users need to understand their requirements so that EdTech solutions are specifically tailored to these needs. Programs like SkillsFuture Singapore matches workforce demands with education outputs.

3 **Increased funding:** This can be achieved through public-private partnerships, donor programs, or venture capital. Just as Development finance institutions (DFIs) have boosted FinTech startups, similar initiatives can foster the rise of Edtech.

4 **Pan-African accelerators:** Africa has only a couple of dedicated EdTech accelerators, among which Injini or iHub. Increasing mentorship, funding, and networking efforts will promote cross-border growth, as shown by initiatives like Google for Startups Africa, which has supported 120+ startups and raised $230M in VC funding. Another example is that of Techstars Africa, which mentors more than 80 startups across 10 countries.

With efficient policies, collaborative networks and sufficient funding, Africa's EdTech sector can scale innovative solutions and transform education for millions.

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